MSc Biostatistics & Health Data Science

Missing Data Analysis - Assignment 1

The file **data.csv** contains data from a random sample of 500 individuals from a study investigating the association of Systolic Blood Pressure with LDL Cholesterol levels. The available variables are the Systolic Blood Pressure levels (SBP; in mmHg), the LDL Cholesterol levels (Chol; in mg/dl), the Age (in years) and the Gender of the participants.

Complete the following tasks:

A. Count the missing values for each variable in the dataset.

B. Calculate appropriate descriptive statistics for all the variables in the dataset, removing missing values where they exist. Calculate the pairwise correlation matrix of Age, Systolic Blood Pressure and LDL Cholesterol, handling missing values with *a)* listwise deletion and *b)* pairwise deletion. Are there differences in the calculations between the two approaches? Where do they stem from?

C. Does the probability of missingness on Systolic Blood Pressure depend on the values of Age? What can we infer about the missingess mechanism on Systolic Blood Pressure?

D. Impute the missing values of Systolic Blood Pressure and Cholesterol with their respective mean values from the observed data (you'd better create new columns in the dataset for the imputed variables and keep the observed ones as is). After the imputation, calculate their mean values and compare them with the ones you obtain from the complete-case analysis.

E. Fit a linear regression model of Systolic Blood Pressure to Age and Gender, based on the complete cases. Using the fitted model, impute the missing values on Systolic Blood Pressure using *a)* the predicted values (from the fitted model) *b)* the predicted values + a random error term drawn from $N\left(0, \widehat{\sigma^2}\right)$, where $\widehat{\sigma^2}$ is the estimated from the fitted model variance of the errors. Compare the mean value of systolic blood pressure and its standard error obtained from the two regression imputation approaches. Comment on the results.

The deliverables for the assignment include a file with your results (including R output) and comments/interpretations, plus your R code for all questions either in a table at the end of the file or in a separate R script. Optionally, instead of the above, you can submit the result (word/pdf) of an Rmarkdown file that combines your R code with the results.